

ROHIT KHED

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CAREER SUMMARY

ECE graduate with internship experience in VLSI design (Verilog, SystemVerilog, RTL-to-physical design flow) and embedded systems/IoT (STM32, Arduino, NodeMCU). Built and verified digital designs, developed GPS- and sensor-based embedded systems, and delivered team projects from concept to working hardware.

TECHNICAL SKILLS

Languages: Embedded C, C++, Verilog, SystemVerilog, HTML/CSS

Microcontrollers & Modules: STM32 (STM32F1/F4), Arduino UNO, NodeMCU (ESP8266), SIM808 GSM/GPS module

EDA & Dev Tools: EDA Playground, OpenRoad, STM32CubeIDE, STM32CubeMX, Arduino IDE

Protocols & Comms: UART/USART, I2C, GPRS/HTTP, Wi-Fi

Other Tools: Microsoft Office, SAP Basics

Languages Spoken: English, Kannada, Hindi | **Interests:** Photography, Trekking, Sports, Volunteering, Reading

WORK EXPERIENCE

VLSI Design Engineer — Intern, Rooman Technologies (with IIT Guwahati & Wadhvani Foundation) 2024 – 2025

- Completed an industry/research internship covering the full VLSI design flow — specification, RTL design, logic synthesis, and physical design (floor planning, placement, routing) — using Verilog and SystemVerilog.
- Designed and simulated an 8-bit bidirectional shifter and a 4-bit up/down counter in Verilog/SystemVerilog, validating functionality with self-written testbenches on EDA Playground.
- Built a UVM block-level verification environment to test and validate RTL design correctness.
- Gained hands-on exposure to OpenRoad for physical design automation and chip layout.
- Completed Wadhvani Foundation's employability skill-development track alongside the technical program, strengthening communication, teamwork, leadership, and time-management skills.

Embedded Systems Intern (Project-Based), Edu Tantr 2025

- Developed a GPS tracking system on an STM32 microcontroller interfaced with a SIM808 GSM/GPS module, using interrupt-driven UART communication for reliable data exchange.
- Implemented GPRS-based HTTP transmission to push live GPS coordinates to a remote web server at fixed intervals, with GPS-fix detection and error handling for robust field operation.
- Configured microcontroller peripherals (USART, NVIC interrupts) via STM32CubeIDE/STM32CubeMX and debugged system behavior through serial console output.

PROJECTS

IoT-Based Patient Health Monitoring System — Team project CODEFIESTA 4.0 (Best Project Award)

- Built a wearable health-monitoring kit using Arduino UNO, ESP8266 Wi-Fi module, pulse and LM35 temperature sensors, with live readouts on OLED/LCD displays.
- Logged health data to ThingSpeak for remote tracking and displayed real-time alerts and wellness tips on-device.

IoT-Based Anti-Theft Detection and Alarm System — Team mini-project NodeMCU

- Designed a low-cost home-security system on NodeMCU (ESP8266) using PIR and ultrasonic sensors for real-time intrusion detection, with LCD and buzzer alerts.
- Integrated Telegram bot notifications for instant remote alerts and system arm/disarm control from anywhere.

EDUCATION

SKSVMACET, Laxmeshwar, Karnataka 2025

B.E., Electronics and Communication Engineering — 73.6%

BASE PU College, Hubli, Karnataka 2021

Pre-University Course (PUC)

Kendriya Vidyalaya No. 2, Hubli, Karnataka 2019

SSLC

ACCOMPLISHMENTS

- Best Project Award — CODEFIESTA 4.0 Hackathon, 2024 (IoT-Based Patient Health Monitoring System).
- Most Innovative Idea — Entrepreneurship Boot Camp, VTU Belagavi, 2024.
- President, E-SPECTRA Association — led department association activities and events.
- First Student Ambassador, Foundation Day, 2024; active volunteer in multiple technical and non-technical events at SKSVMACET.